

Daily Review

Booklet p.12-14 – Satellites, and p.5 – $W = mg$

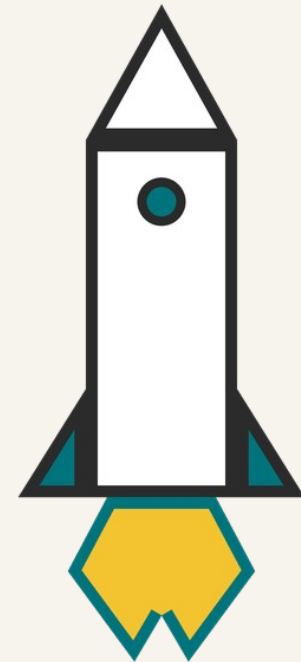
- 1 State the period of a geostationary satellite. 24 hours
- 2 State the altitude of a geostationary satellite. 36 000 km
- 3 What happens to the period as altitude increases? It increases
- 4 A 40 kg crate is on Earth. Calculate its weight. 392 N
- 5 Give two benefits of satellites. GPS; weather forecasts

SPACE

Forces on Rockets

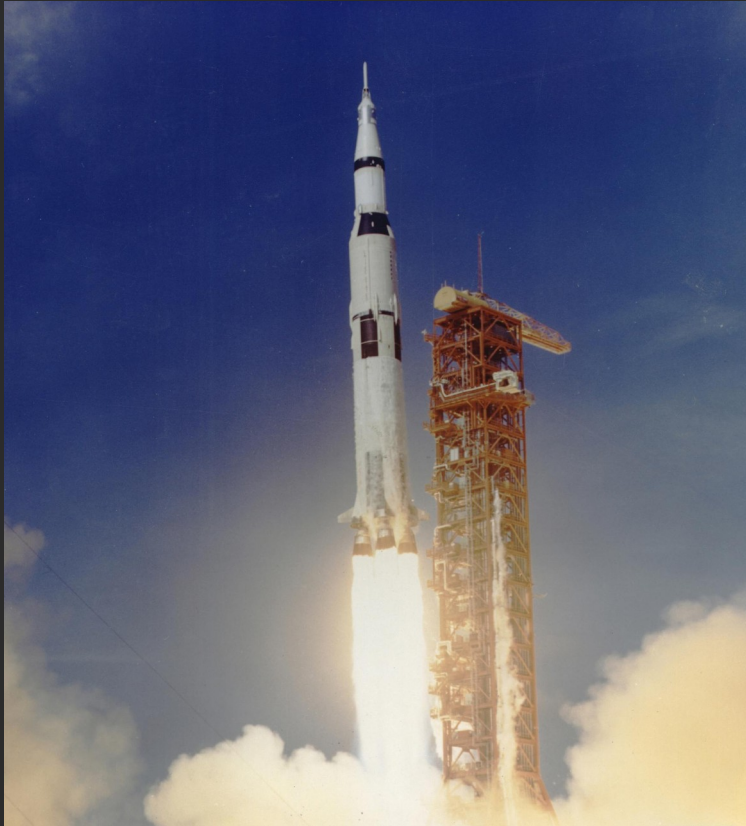
What it takes to leave the launch pad
— and why it works.

National 5 Physics (S3)



THE BIG QUESTION

What does it take to lift a rocket this big off the pad — and make it speed up?



Saturn V — NASA



Artemis II, 2026 — NASA/Michael DeMocker

PART 1

Newton's second law at lift-off

What makes a rocket speed up as it leaves the pad — and by how much.

By the end of this part...

- 1 I can state Newton's second law of motion
- 2 I can apply Newton's second law to problems with rockets

Show me — Describe each word

SHOW ME BOARDS

1 force

A push or a pull, in newtons (N)

2 thrust

The upward force from the engines

3 weight

The force due to gravity on an object

4 air resistance

Friction with the air, acting downwards

5 unbalanced force

What is left when forces do not balance

Show me — Describe each word

SHOW ME BOARDS

6 acceleration

How quickly velocity changes, in m s^{-2}

7 Newton's first law

Balanced forces: still, or constant velocity

8 Newton's second law

An unbalanced force causes acceleration

9 Newton's third law

If A pushes B, B pushes back on A

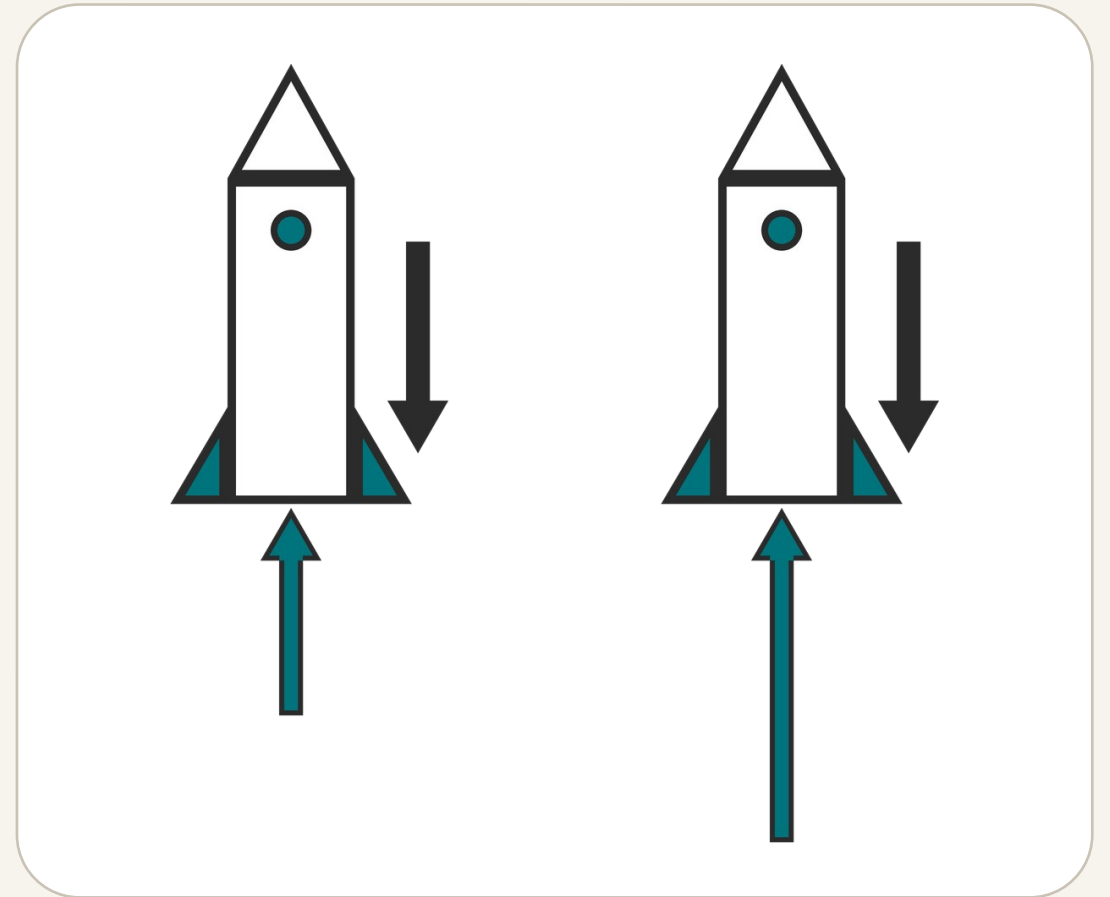
10 Newton pair

Equal, opposite forces on two objects

Balanced, then unbalanced

Booklet p.15 – Newton's first and second laws

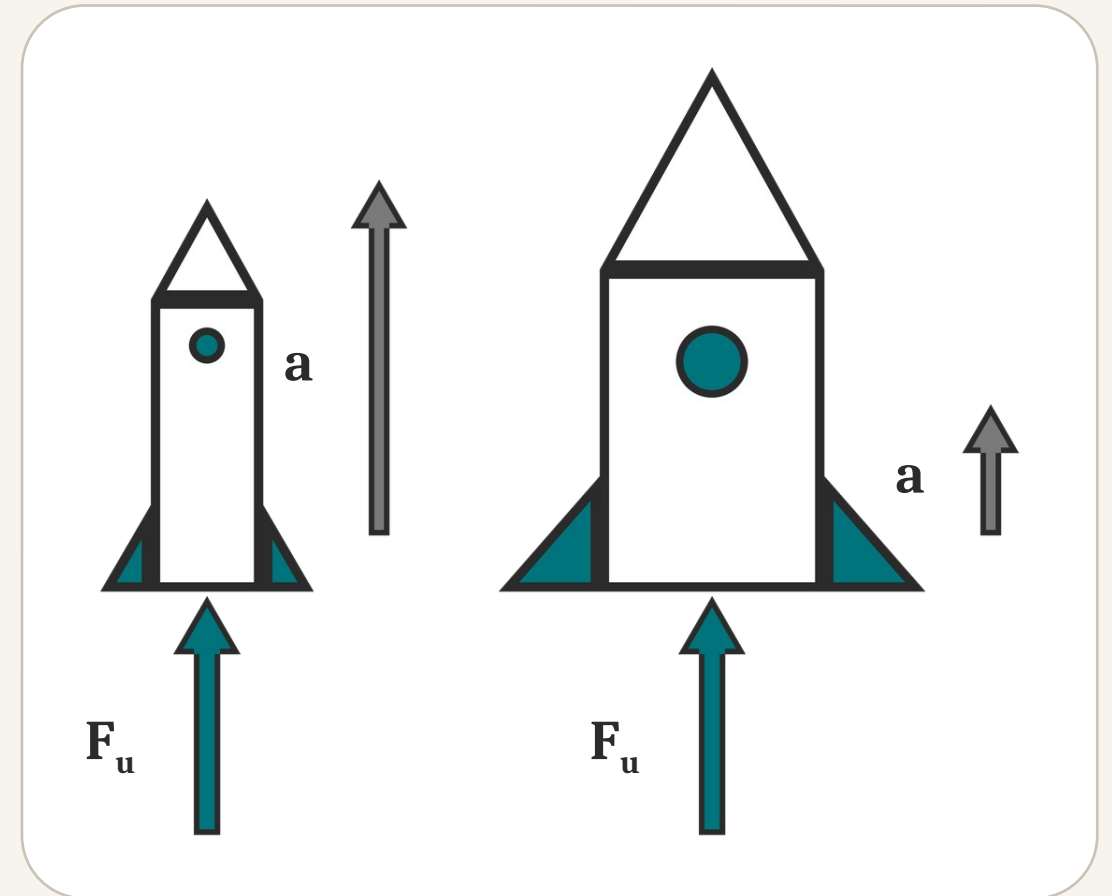
- Balanced forces: a constant velocity, or staying still.
- Thrust bigger than weight: an unbalanced force upwards.
- An unbalanced force makes the rocket accelerate.



Force, mass and acceleration

Booklet p.15 – Newton's second law

- The same unbalanced force acts on both rockets.
- The rocket with less mass has the bigger acceleration.
- Burning fuel cuts a rocket's mass, so its acceleration grows.



Show me — Newton's laws

SHOW ME BOARDS

- 1 The forces on a rocket are balanced. What can it be doing?
- 2 What does an unbalanced force do to an object?
- 3 A rocket's thrust is bigger than its weight. What happens?
- 4 Thrust equals weight. Is the rocket accelerating?
- 5 Which law says an unbalanced force causes acceleration?

Staying still, or constant velocity

Makes it accelerate

It accelerates upwards

No: the forces are balanced

Newton's second law

Show me — Newton's laws

SHOW ME BOARDS

EXTRA PRACTICE

- 1 A lift rises at a steady speed. Are the forces balanced?
- 2 A rocket's thrust is less than its weight. What happens?
- 3 Which law is about balanced forces?
- 4 Two rockets have the same F_u . Which has the bigger a ?
- 5 A car's driving force equals its friction. Describe its motion.

Yes: constant velocity

It stays on the pad

Newton's first law

The one with less mass

Constant velocity

Task 1 – the first two laws

Booklet p.15

Fill in the blanks for Newton's first
and second laws.

Booklet p.15, Task 1

Task 1 – Answers

ANSWERS

Booklet p.15 – first and second laws

Law	The missing words, in order
First law	forces ... velocity
Second law	unbalanced

The relationship

Booklet p.15 – Task 2

$$F_u = m a$$

F_u

unbalanced force,
in newtons (N)

m

mass, in kilograms
(kg)

a

acceleration, in m s^{-2}

Task 2 – label the equation

Booklet p.15

Label each letter in $F_u = ma$
with its quantity and its unit.

Booklet p.15, Task 2

Task 2 – Answers

ANSWERS

Booklet p.15

Letter	Quantity	Unit
F_u	unbalanced force	newton (N)
m	mass	kilogram (kg)
a	acceleration	m s^{-2}

Task 3 – forces at take-off

Booklet p.15

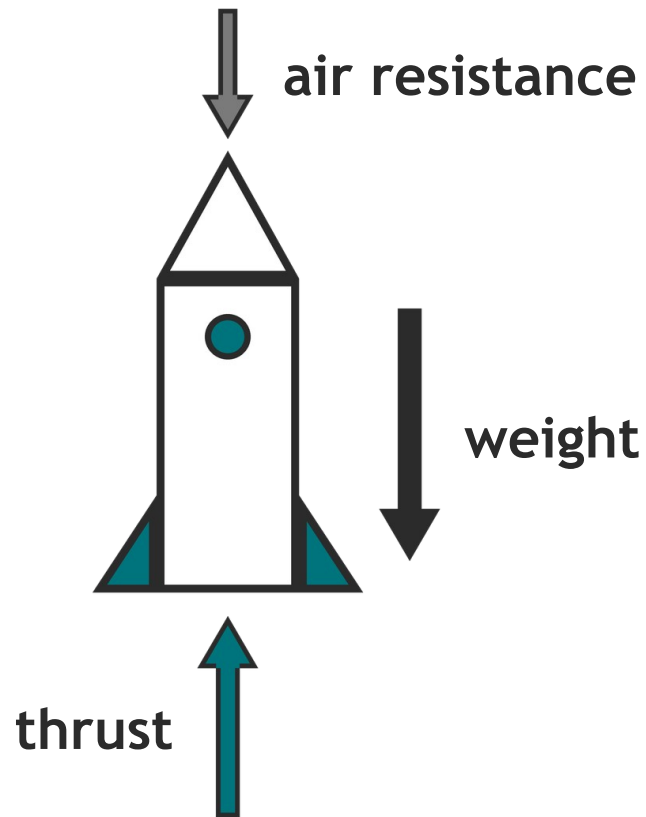
Label all the forces on the rocket at take-off. Word bank: thrust, weight, air resistance, unbalanced force.

Booklet p.15, Task 3

Task 3 – Answers

ANSWERS

Booklet p.15



- Thrust acts upwards.
- Weight and air resistance act downwards.
- Unbalanced force = thrust – weight – air resistance.

Worked example 1

Lift-off, step 1 – the unbalanced force

A rocket of mass 2400 kg has a thrust of 36 000 N at lift-off.
Air resistance can be ignored. Find the **unbalanced force**.

$$m = 2400 \text{ kg}$$

$$g = 9.8 \text{ N kg}^{-1}$$

$$\text{thrust} = 36\,000 \text{ N}$$

$$F_u = ?$$

$$W = m g$$

$$W = 2400 \times 9.8$$

$$W = 23\,520 \text{ N}$$

$$F_u = 36\,000 - 23\,520$$

$$\underline{F_u = 12\,480 \text{ N upwards}}$$

Worked example 1 – the acceleration

The same rocket: mass 2400 kg, unbalanced force 12 480 N.
Calculate its initial **acceleration**.

$$F_u = 12\,480\text{ N}$$

$$m = 2400\text{ kg}$$

$$a = ?$$

$$F_u = m a$$

$$a = F_u/m$$

$$a = 12\,480/2400$$

$$\underline{a = 5.2\text{ m s}^{-2}}$$

Your turn 1 – lift-off

SHOW ME BOARDS

A rocket of mass 1500 kg has a thrust of 27 000 N at lift-off.
Air resistance can be ignored. Calculate its initial acceleration.

$$m = 1500 \text{ kg}$$

$$g = 9.8 \text{ N kg}^{-1}$$

$$\text{thrust} = 27\,000 \text{ N}$$

$$a = ?$$

$$14\,700 \text{ N}; \quad 12\,300 \text{ N}$$
$$a = 8.2 \text{ m s}^{-2}$$

$$W = mg = 1500 \times 9.8 = \dots\dots \text{ N}$$

$$F_u = 27\,000 - \dots\dots = \dots\dots \text{ N}$$

$$F_u = m a$$

$$a = \dots\dots/1500$$

$$\underline{a = \dots\dots \text{ m s}^{-2}}$$

Worked example 2 – landing on Mars

A 350 kg space vehicle is falling towards the surface of Mars.
Its engines give 2200 N upwards. Find its acceleration.

$$m = 350 \text{ kg}$$

$$g = 3.7 \text{ N kg}^{-1}$$

$$\text{thrust} = 2200 \text{ N}$$

$$a = ?$$

$$W = mg = 350 \times 3.7 = 1295 \text{ N}$$

$$F_u = 2200 - 1295 = 905 \text{ N}$$

$$F_u = m a$$

$$a = 905/350$$

$$\underline{a = 2.6 \text{ m s}^{-2} \text{ upwards}}$$

Your turn 2 – landing on the Moon

SHOW ME BOARDS

A 600 kg lander is falling towards the surface of the Moon.
Its engines give 1500 N upwards. Find its acceleration.

$$m = 600 \text{ kg}$$

$$g = 1.6 \text{ N kg}^{-1}$$

$$\text{thrust} = 1500 \text{ N}$$

$$a = ?$$

$$\begin{aligned} &960 \text{ N}; 540 \text{ N} \\ &a = 0.90 \text{ m s}^{-2} \text{ upwards} \end{aligned}$$

$$W = mg = 600 \times \dots = \dots \text{ N}$$

$$F_u = \dots - \dots = \dots \text{ N}$$

$$F_u = m a$$

$$a = \dots / \dots$$

$$\underline{a = \dots \text{ m s}^{-2}, \text{ direction } \dots}$$

Hinge question

FINGER VOTE

Think first — then show A to E on your fingers, 1 to 5

A rocket of mass 4000 kg has a thrust of 51 200 N at lift-off.
Air resistance can be ignored. Its initial acceleration is

- A 0.33 m s^{-2}
- B 3.0 m s^{-2}
- C 9.8 m s^{-2}
- D 12.8 m s^{-2}
- E 22.6 m s^{-2} .

Answer B

Show me — Newton's second law

SHOW ME BOARDS

EXTRA PRACTICE

1

A 1000 kg rocket: thrust 16 000 N.
Calculate its weight on Earth.

9800 N

2

...the unbalanced force at lift-off.

6200 N

3

...and its initial acceleration.

6.2 m s^{-2}

4

Why does a rocket's acceleration increase as it burns fuel?

Its mass falls, so a grows

5

A 50 kg probe accelerates at 0.20 m s^{-2} .
Find the unbalanced force.

10 N

Watch out

THE MISTAKE

- Putting the thrust straight into $F_u = ma$
- Leaving the weight out at lift-off

THE CORRECT ANSWER

- Find $F_u = \text{thrust} - \text{weight}$ first
- Weight acts down: take it away from the thrust

Task 4 – try the example

Booklet p.16

A rocket: 1850 kg, thrust 30 kN.
Complete the force diagram, then
calculate the initial acceleration.

Booklet p.16, Task 4

Task 4 – Answers

ANSWERS

Booklet p.16

Step	Working	Answer
Weight	$W = mg = 1850 \times 9.8$	18 130 N
F_u	$F_u = 30\,000 - 18\,130$	11 870 N
Acceleration	$a = F_u \div m = 11\,870 \div 1850$	6.4 m s^{-2}

Daily Review

Booklet p.15 — Part 1, and p.5 — $W = mg$

- 1 What does an unbalanced force do to an object?
Makes it accelerate
- 2 A 5.0 kg mass has an unbalanced force of 15 N. Calculate a .
 3.0 m s^{-2}
- 3 Name the force that acts upwards on a rocket at take-off.
Thrust
- 4 A 70 kg astronaut is on the Moon. Calculate the weight.
112 N
- 5 State Newton's second law.
An unbalanced force causes acceleration

PART 2

Newton pairs, and Review 1

Every force has a partner — and that is how a rocket flies.

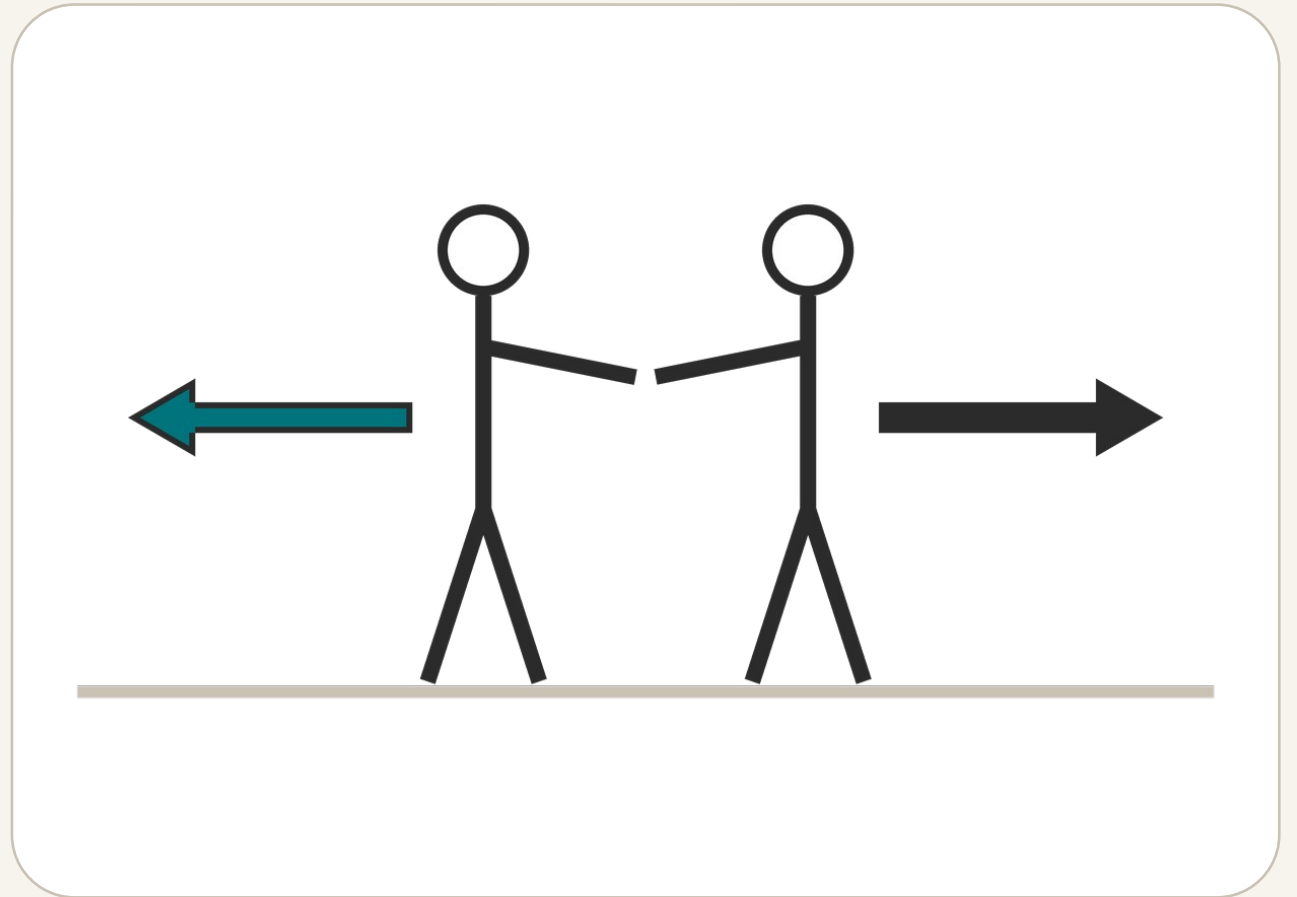
By the end of this part...

- 1 I can state Newton's third law of motion
- 2 I can apply Newton's third law to problems with rockets

Newton's third law

Booklet p.15

- If A pushes on B, then B pushes back on A.
- The forces are equal in size, opposite in direction.
- Each force acts on a different object.



Task 1 – the third law

Booklet p.15

Fill in the blanks for Newton's
third law.

Booklet p.15, Task 1

Task 1 – Answers

ANSWERS

Booklet p.15 – third law

Law	The missing words, in order
Third law	force ... b ... opposite ... a

Agree, disagree or it depends?

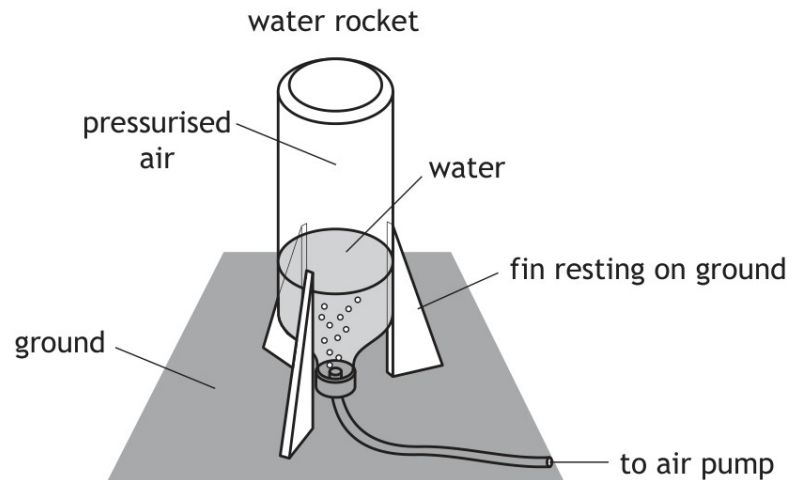
SHOW ME BOARDS

Boards up: AGREE, DISAGREE or IT DEPENDS — and why

- 1 Gases push on the ground, and that lifts the rocket.
Disagree: they push on the rocket
- 2 The force on the rocket and the force on the gases cancel out.
Disagree: different objects
- 3 The gases push on the rocket as hard as it pushes on them.
Agree: equal and opposite
- 4 The rocket pushes harder on the gases than they push back.
Disagree: the forces are equal
- 5 Thrust beats weight, so it speeds up.
It depends: air resistance too

Show me – the water rocket

SHOW ME BOARDS



1

What does the bottle push downwards?

The water

2

What pushes the bottle upwards?

The water, pushing back

3

How do the two forces compare?

Equal and opposite

4

What must the thrust be bigger than?

The weight

Demo – the water rocket

DEMO

Predict: which way will the water go?

Predict: which way will the bottle go?

Watch the launch.

Explain it as a Newton pair: what pushes on what?

Task 5 – the Newton pair

Booklet p.16

Complete the diagram to show the Newton pair acting on the rocket.

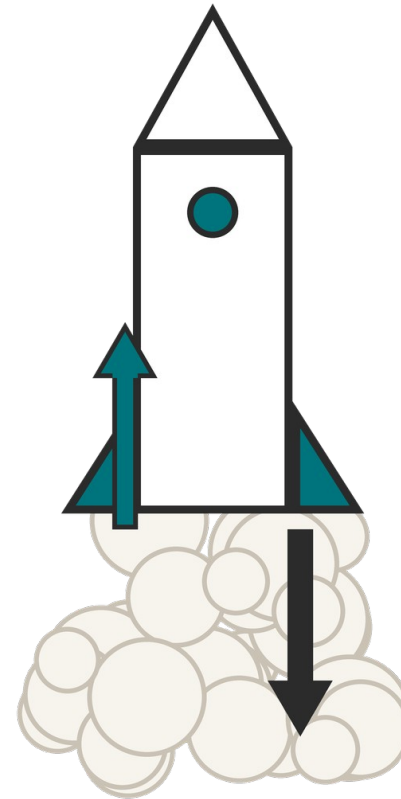
Booklet p.16, Task 5

Task 5 – Answers

ANSWERS

Booklet p.16

- Up: force on the rocket, due to the gases.
- Down: force on the gases, due to the rocket.
- Equal in size, opposite in direction.



A real engine — the RS-25

Newton's third law at work



- The engine pushes hot gases out.
- The gases push back on the engine: equal and opposite.
- No ground or air is needed, so it works in space.
- Four RS-25s power the core stage of Artemis II's rocket.

Show me — Newton pairs

SHOW ME BOARDS

EXTRA PRACTICE

- 1 You push on a wall. What pushes back on you?
- 2 An astronaut throws a tool forwards. What happens to them?
- 3 A rocket pushes 8000 N on its gases. Find the force on the rocket.
- 4 Why do the two forces in a Newton pair not cancel out?
- 5 State Newton's third law.

The wall

They move backwards

8000 N, upwards

They act on different
objects

Equal, opposite forces
on each other

Checkpoint

1 State Newton's second law.

An unbalanced force
causes acceleration

2 A 3000 kg rocket has a thrust of 44 400 N. Find its acceleration.

5.0 m s^{-2}

3 Use Newton's third law to explain how a rocket lifts off.

Gases pushed down;
rocket pushed up

Command words – 1 of 2

Word	What it asks for	A Space example
State	Name it, or give it in brief form.	State Newton's third law.
Describe	Give its features: what happens.	Describe the motion of a rocket with balanced forces.
Explain	Link cause and effect.	Explain how a rocket lifts off.

Command words – 2 of 2

Word	What it asks for	A Space example
Suggest	Use your physics in a new situation.	Suggest why a lander fires its engines before landing.
Predict	Say what may happen, from the facts given.	Predict how a changes as a rocket burns fuel.
Justify	Give reasons that back up your answer.	Is the rocket accelerating? Justify your answer.

What a 3-mark answer shows

Open-ended questions

- 3 marks: good understanding – the principles, a relationship or equation, applied. It need not be complete.
- 2 marks: reasonable understanding – relevant statements.
- 1 mark: limited understanding – a little of the physics.
- 0 marks: nothing relevant, or just repeating the question.

Open-ended practice

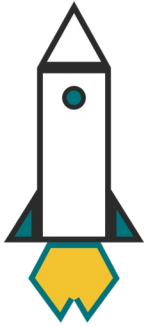
In your jotter (3 marks)

Artemis II's crew left Earth on a rocket and came home under parachutes.

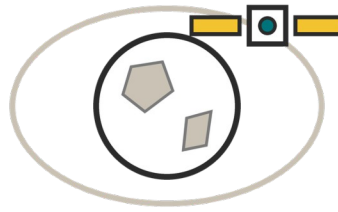
Using your knowledge of physics, comment on the forces acting on the spacecraft during its journey.



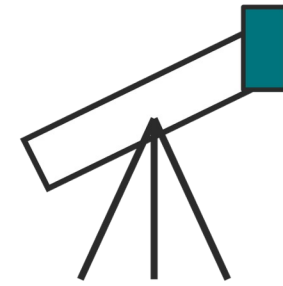
Careers – where this physics goes



Rocket engineer



Satellite engineer



Astronomer

- UKube-1, Scotland's first satellite, was built in Glasgow.
- SaxaVord Spaceport on Unst, Shetland, is licensed for launches.

Exam practice

Past paper and specimen paper

- N5 Physics 2025, Section 2, Q2(a). 7 marks — a rocket's weight and acceleration at launch.
- N5 Physics specimen paper, Section 1, Q4. 1 mark — a Newton pair.
- Specimen paper, Section 2, Q2(a) and (c). 6 marks — a spacecraft leaving the ISS, and its parachutes.
- Then booklet p.17, Task 6.

Review 1

Booklet p.18

Check your booklet is up to date.
Test yourself on the definitions.
Do the Space practice questions on the
website, Parts A and B, and mark them.
Fill in your learning log.

Booklet p.18, Review

LOOKING AHEAD

The Big Bang Theory

How did the universe begin — and how do we know how old it is?

Plenary

- 1 The big question: what does it take to lift a rocket off the pad?
- 2 State Newton's second law.
- 3 State Newton's third law.
- 4 Name the forces on a rocket at take-off.
- 5 A 200 kg probe has an unbalanced force of 50 N. Calculate a .

Thrust bigger than weight

An unbalanced force causes acceleration

If A pushes B, B pushes back on A

Thrust, weight, air resistance

0.25 m s^{-2}

Plenary

EXTRA PRACTICE

- 1 What is a Newton pair?
- 2 What must be true of the thrust for a rocket to lift off?
- 3 Write down the relationship for unbalanced force.
- 4 What is the unit of acceleration?
- 5 A 10 kg mass accelerates at 4.0 m s^{-2} . Find the unbalanced force.

Equal, opposite forces
on two objects

Bigger than the weight

$$F_u = ma$$

$$\text{m s}^{-2}$$

$$40 \text{ N}$$

Extension

- 1 A rocket's thrust stays the same as it burns fuel. Explain what happens to a.
- 2 Explain why a rocket works in space, where there is no air to push on.
- 3 A 20 000 kg rocket lifts off Earth at 1.2 m s^{-2} . Find the thrust.

Extension

EXTRA PRACTICE

- 1 Why is a lander's descent speed controlled with an upward thrust?
- 2 Explain how a probe in deep space can slow down.
- 3 A 900 kg lander on the Moon has a thrust of 2340 N. Find a .

Course specification – 1 of 2

N5 Physics course specification v4.0, p.8 – Space exploration

- Knowledge of Newton's second and third laws and their application to space travel, rocket launch and landing.

Course specification – 2 of 2

N5 Physics course specification v4.0, p.6 – Newton's laws

- Application of Newton's laws and unbalanced forces to explain and/or determine acceleration for situations where more than one force is acting.
- Use of an appropriate relationship to solve problems involving unbalanced force, mass and acceleration for situations where one or more forces are acting in one dimension or at right angles.
- $F = ma$
- Explanation of motion resulting from a 'reaction' force in terms of Newton's third law.